

Wave Speed Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links wave speed (v), frequency (f), and wavelength (λ). Cover the unknown to read the formula: $v = f \times \lambda$, $f = v \div \lambda$, and $\lambda = v \div f$.

Round 1 — Retrieve It

1. What three quantities does the wave speed triangle connect?

2. Write the formula for wave speed.

3. Write the formula for frequency.

4. Which variable do you cover to find wavelength, and what formula results?

Round 2 — Sketch It

Draw the wave speed triangle. Put wave speed (v) on top; put frequency (f) and wavelength (λ) on the bottom. Circle the two quantities you would multiply to find wave speed.

Wave Speed — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A wave has a frequency of 5 Hz and a wavelength of 4 m. What is its wave speed?

2. A wave travels at 60 m/s with a wavelength of 3 m. What is its frequency?

3. A wave travels at 45 m/s with a frequency of 9 Hz. What is its wavelength (λ)?

Round 4 — Reason from Evidence

Two waves travel through the same medium at a constant wave speed of 300 m/s. Wave A has a frequency of 100 Hz; Wave B has a frequency of 150 Hz. Use the triangle to find each wave's wavelength, then say what happens to wavelength as frequency increases and how you know.

Wave Speed — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain how the wave speed triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C

Connect

Connect wave speed to the relationship between frequency and wavelength. When wave speed stays constant, what happens to wavelength as frequency increases, and how does that relate to $v = f \times \lambda$?

E

Extend

Create your own problem that asks for wavelength (λ). Give the numbers and show the work using $\lambda = v \div f$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Wave Speed

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Wave speed, frequency, and wavelength.
2. $v = f \times \lambda$.
3. $f = v \div \lambda$.
4. Cover λ ; $\lambda = v \div f$.

Sketch It — look for

Wave speed (v) on top, with frequency (f) and wavelength (λ) on the bottom. Wave speed is found by multiplying frequency $\times$ wavelength (the two bottom quantities).

Apply It

1. $v = f \times \lambda = 5 \times 4 = \mathbf{20 \text{ m/s}}$.
2. $f = v \div \lambda = 60 \div 3 = \mathbf{20 \text{ Hz}}$.
3. $\lambda = v \div f = 45 \div 9 = \mathbf{5 \text{ m}}$.

Reason from Evidence — look for

Wave A: $\lambda = v \div f = 300 \div 100 = 3 \text{ m}$. Wave B: $\lambda = v \div f = 300 \div 150 = 2 \text{ m}$. When wave speed is constant, wavelength decreases as frequency increases — they are inversely related (f and λ are the two bottom quantities that multiply to a fixed v).

ACE — Extend look-for

Many answers possible. Check that the problem asks for wavelength and the work correctly uses $\lambda = v \div f$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.